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Do Volatility-Managed Portfolios Survive Out of Sample? A Pre-Registered Replication and Real-Time Test of Moreira and Muir (2017)

A pre-registered replication and out-of-sample study using the Kenneth French Data Library

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Abstract

Volatility-managed portfolios, which reduce exposure to a factor after volatile months and increase it after calm months, were reported by Moreira and Muir (2017) to earn large abnormal returns across most equity factors. Motivated by the argument of Ioannidis (2005) that most published research findings are false, this study subjects the result to a pre-registered replication and out-of-sample test. The hypotheses, tests, decision rules, a power analysis and a grid of 288 robustness specifications were fixed before any data were downloaded. Using the free Kenneth French Data Library, the original results are reproduced almost exactly, with every alpha within half a standard error of its published value. On the 127 months from January 2016 to July 2026, however, no spanning alpha remains significantly positive, and the alpha of managed momentum falls from 12.80% to 0.51% per year, a decline that survives, narrowly, a post-hoc adjustment for the smaller positions the strategy held after 2016. The pre-registered rule also classifies the alphas of profitability and value as significantly smaller, but on a risk-adjusted basis these declines are not significant. For an investor implementing the strategy in real time, the pre-registered primary test, the Sharpe ratio gain of managed momentum after 10 basis points of trading costs, is +0.03 (95% confidence interval −0.32 to +0.39, p = 0.45). The point estimates for the other five factors are negative, but no Sharpe ratio gain is significantly below its pre-2016 value, and the primary test has a power of only about 18% against a gain of the pre-2016 size. At the same time, 26 of the 48 momentum specifications appear significant by the pre-registered p-value, all with longer variance windows than pre-registered, although they amount to four distinct strategies and only 10 have a 95% confidence interval that excludes zero. This shows how choices made after seeing the data could have produced a spurious confirmation. The results suggest that the historical advantage of volatility management was concentrated in rare crash episodes and could only partly be captured in real time.

Keywords: volatility-managed portfolios, momentum, replication, out-of-sample testing, pre-registration, multiple testing, factor investing

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1. Introduction

In 2017, *The Journal of Finance* published a result that seemed almost too good to be true. Moreira and Muir (2017) showed that an investor who simply reduced exposure to an equity factor after a volatile month, and increased it after a calm month, would have earned large and highly significant abnormal returns across almost every major factor. For the momentum factor, the alpha of this volatility-managed strategy reached 12.51% per year with a t-statistic above seven, and for the market portfolio the strategy produced an annualised alpha of 4.86%. The rule requires nothing more than a month of daily returns and a division, and it was presented as a challenge to standard models of the risk–return trade-off. Since its publication, the paper has become one of the most discussed contributions to empirical asset pricing, and volatility targeting has become a standard feature of commercial investment products.

A decade earlier, and in a completely different discipline, Ioannidis (2005) had published a paper with a deliberately provocative title: "Why most published research findings are false". His argument was not that researchers are dishonest, but that the probability that a statistically significant finding reflects a true relationship depends on three quantities that are rarely reported: the prior odds that the hypothesis is true, the statistical power of the test, and the degree of flexibility or bias in how the analysis was conducted. When many hypotheses are tested, effects are small, designs are flexible and many teams study the same data, a large share of "discoveries" will be false positives, even if every individual study is conducted in good faith. Ioannidis (2005) wrote about medicine, but each of his warning signs describes empirical finance with uncomfortable precision. Hundreds of return predictors have been published (Harvey et al., 2016), monthly effect sizes are small relative to the noise in returns, researchers choose freely among sample periods, weighting schemes and parameters, and almost everyone works with the same CRSP and Compustat databases.

The present study takes Ioannidis's diagnosis as a working method rather than as a slogan. It asks whether the volatility-managed portfolio result of Moreira and Muir (2017) survives three increasingly demanding tests. The first is replication: can the published numbers be reproduced from free, public data? The second is an out-of-sample test on the ten and a half years of returns, from January 2016 to July 2026, that the original authors could not have observed. The third is implementability: does the strategy still deliver a higher Sharpe ratio when an investor runs it in real time, using only information available at each date, and pays trading costs? Crucially, every hypothesis, test, decision rule and robustness check was written into a pre-registration plan before any data were downloaded, so that the results could not shape the questions.

This design responds directly to a debate that has emerged since 2017. Cederburg et al. (2020) show, across 103 equity strategies, that the spanning-regression alphas used by Moreira and Muir (2017) do not translate into better performance for an investor who must implement the strategy in real time. Barroso and Detzel (2021) find that, once trading costs are included,

volatility management of factors other than the market generally produces zero abnormal returns and lower Sharpe ratios. DeMiguel et al. (2024), in contrast, argue that a conditional multifactor version of the strategy still outperforms out of sample after costs. What is missing from this literature is a test on genuinely new data, conducted under a protocol fixed in advance, and evaluated with decision rules that distinguish between an effect that has disappeared and an effect that the data are simply too short to detect.

The contribution of this study is threefold. First, it provides an exact replication of the original results from the freely available Kenneth French Data Library, matching the published observation counts for every factor and reproducing every published alpha to within half a standard error. Second, it reports a pre-registered out-of-sample test with an explicit power analysis. Because the power of the market-factor test was estimated at only 29% before any data were seen, the primary test was moved to the momentum factor, where power was estimated at 88%. That estimate, however, concerned the spanning alpha, and Section 3.6 shows that the primary test, a comparison of real-time Sharpe ratios, turned out to have far less power. Third, it documents the full distribution of results across 288 pre-specified robustness specifications, which makes it possible to show how much a researcher could have changed the conclusion simply by choosing a different, equally defensible specification after seeing the data.

The results are sobering. The replication succeeds almost perfectly. On the new data, however, the spanning alpha of managed momentum falls from 12.80% to 0.51% per year, and the pre-registered decision rule classifies the estimates for momentum, profitability and value as significantly smaller than their in-sample counterparts. Only the momentum decline survives, narrowly, a post-hoc adjustment for the much smaller positions the managed portfolios held after 2016. The primary pre-registered test, the real-time Sharpe ratio gain of managed momentum after 10 basis points of trading costs, is +0.03 with a 95% confidence interval from −0.32 to +0.39 and a p-value of 0.45. For the market, size, value and investment factors, the point estimates are negative, but none of these results is labelled *Contradicted*. The primary test also has low power, about 18% against a gain of the size achieved before 2016. At the same time, 26 of the 48 momentum specifications in the robustness grid are "significant" at the 5% level by the pre-registered p-value, although they amount to only four distinct strategies and only 10 have a 95% confidence interval that excludes zero. All of them use longer variance windows than the primary specification. A researcher free to choose among them could have written a very different paper.

The remainder of the paper is organised as follows. Chapter 2 reviews the literature on false discoveries in science and finance, on factor models and momentum, and on volatility-managed portfolios and their critiques. Chapter 3 describes the pre-registered research design, the data, the construction of the strategies and the statistical tests. Chapter 4 presents the replication, the out-of-sample results and the robustness grid. Chapter 5 interprets the findings in light of Ioannidis (2005) and discusses the limitations of the study, and Chapter 6 concludes.

2. Literature Review

This chapter reviews the academic literature on which the present study builds. It first examines the general argument that many published findings are false and the tools proposed to address it, then turns to the specific evidence of data mining and replication failure in empirical asset pricing. It subsequently reviews the factor models and the momentum anomaly on which volatility management is applied, the original evidence on volatility-managed portfolios, and the critiques that have followed. The chapter concludes by identifying the research gap addressed by this study.

2.1 Why most published research findings are false

The starting point of this study is the framework of Ioannidis (2005), who modelled the research process as a sequence of hypothesis tests and asked what fraction of statistically significant results should be expected to be true. He defined the positive predictive value (PPV) of a claimed finding as the probability that a relationship declared significant is in fact real. In the simplest version of his model, without bias, the PPV depends on the pre-study odds R that a tested relationship is true, on the significance level α, and on the statistical power 1 − β of the test:

$$\text{PPV} = (1 - \beta)R / (R - \beta R + \alpha)$$

The implication is immediate. If the prior odds that a tested relationship is real are one to ten (R = 0.1, so that about one tested relationship in eleven is real) and studies have 80% power at the 5% level, a significant finding is true with a probability of about 62%. If power falls, or if the prior odds are lower, the PPV falls below one half, and a significant result becomes more likely to be false than true. Ioannidis (2005) then introduced a bias term u, the share of analyses that would not have produced a positive result but are nonetheless reported as positive because of flexible design, selective reporting or analytical choices made after seeing the data. Even moderate bias sharply reduces the PPV.

From this model, Ioannidis (2005) derived six corollaries. A research finding is less likely to be true when the studies conducted in a field are smaller, when effect sizes are smaller, when a greater number of relationships is tested with less pre-selection, when there is greater flexibility in designs, definitions, outcomes and analytical modes, when there are greater financial and other interests and prejudices, and when more teams are involved in chasing statistical significance in the same "hot" field. These corollaries provide the organising structure for the design of the present study, which is set out in Chapter 3.

Simmons et al. (2011) demonstrated the practical force of the fourth corollary. In a series of simulations and experiments, they showed that undisclosed flexibility in data collection and analysis, such as the choice of when to stop collecting data, which control variables to include and which observations to exclude, allows researchers to present almost any hypothesis as statistically significant. Their proposed remedy was transparency about every decision. Nosek

et al. (2018) went further and argued for pre-registration: specifying the hypotheses, the data, the tests and the decision rules before the outcome is known. Pre-registration does not prevent exploratory analysis, but it makes the distinction between prediction, which tests a hypothesis, and postdiction, which generates one, visible to the reader. The present study adopts this approach.

2.2 Data mining and replication in empirical asset pricing

Empirical finance is particularly exposed to the problems identified by Ioannidis (2005), because its central dataset of historical stock returns is fixed, shared by all researchers, and cannot be extended by running a new experiment. Every new hypothesis is tested on data that have already been used to test many others. Harvey et al. (2016) documented the scale of this problem by cataloguing hundreds of factors proposed in the literature to explain the cross-section of expected returns. Accounting for the number of tests that have been run, and for those that were run but never published, they argued that the conventional t-statistic threshold of 2.0 is far too permissive, and that by 2012 a newly proposed factor needed a t-statistic of at least 3.0 to be considered credible.

The practical consequences of data mining appear clearly in the post-publication performance of anomalies. McLean and Pontiff (2016) studied 97 characteristics shown in published papers to predict stock returns and compared their returns in three periods: the original sample, the period between the end of the sample and publication, and the period after publication. Portfolio returns were 26% lower out of sample and 58% lower after publication. They interpreted the first decline as the signature of statistical bias in the original studies and the additional post-publication decline as the effect of investors learning from academic research and trading on it. Both mechanisms imply that a published in-sample estimate is a biased guide to the future.

Replication studies have reached conflicting conclusions about how serious the problem is. Hou et al. (2020) attempted to replicate 452 anomalies with a common methodology and found that 65% failed to reach the conventional significance threshold of a t-statistic of 1.96, and 82% failed a multiple-testing hurdle of 2.78. Jensen et al. (2023), using a Bayesian framework and a global dataset covering 93 countries, reached a more optimistic conclusion: the majority of asset pricing factors can be replicated, can be grouped into 13 themes, and are robust across markets. The difference between the two studies illustrates a point central to this paper: whether a finding "replicates" depends heavily on how replication is defined, on the statistical standard applied, and on whether the question is statistical significance or economic value to an investor.

2.3 Factor models and the momentum anomaly

The factors used in this study come from the asset pricing models that dominate the academic literature. Fama and French (1993) proposed a three-factor model in which the excess

return on the market (Mkt) is complemented by a size factor (SMB, small minus big) and a value factor (HML, high minus low book-to-market). Fama and French (2015) later added a profitability factor (RMW, robust minus weak operating profitability) and an investment factor (CMA, conservative minus aggressive asset growth). All five factors are constructed as long–short or excess returns and are published, with daily and monthly returns, in the Kenneth French Data Library.

The momentum factor has a different origin. Jegadeesh and Titman (1993) showed that stocks with high returns over the previous three to twelve months continue to outperform stocks with low past returns over the following months. Carhart (1997) added a momentum factor to the three-factor model to evaluate mutual fund performance, and momentum has since become one of the most studied anomalies in finance. Its defining feature, beyond its high average return, is its exposure to rare but severe crashes. Daniel and Moskowitz (2016) showed that momentum strategies experience infrequent and persistent strings of large negative returns, that these crashes occur in "panic states" following market declines when volatility is high, and that they are partly forecastable. In the data used here, the worst monthly momentum return is −52.61%, recorded in August 1932.

This crash risk explains why momentum is the natural testing ground for volatility management. Barroso and Santa-Clara (2015) showed that the risk of momentum is highly variable over time and predictable from its own recent realised volatility, and that scaling the strategy to a constant volatility target virtually eliminates crashes and nearly doubles its Sharpe ratio. Daniel and Moskowitz (2016) similarly reported that a dynamic momentum strategy based on forecasts of its mean and variance approximately doubles both alpha and Sharpe ratio. In other words, the momentum result of Moreira and Muir (2017) was not isolated: it confirmed earlier findings on the same factor.

2.4 Volatility-managed portfolios

Moreira and Muir (2017) generalised the idea of risk-managed momentum to a broad set of factors. Their strategy scales the monthly return of a factor by the inverse of its realised variance in the previous month, measured from daily returns. A constant c is chosen so that the managed portfolio has the same unconditional volatility as the original factor. The strategy therefore takes more risk when recent volatility has been low and less when it has been high. The central evidence comes from spanning regressions of the managed factor return on the unmanaged one. A positive intercept indicates that combining the managed and unmanaged portfolios expands the mean-variance frontier available to an investor who already holds the factor.

The reported alphas were economically large and statistically strong. For the market, the alpha was 4.86% per year with a standard error of 1.56 and an appraisal ratio of 0.34. For momentum, the alpha was 12.51% with a standard error of 1.71 and an appraisal ratio of 0.875. Alphas were also positive and significant for profitability (2.44%) and for several other factors

outside the Fama–French set, while size (−0.58%) and investment (0.38%) produced small and insignificant alphas. Moreira and Muir (2017) argued that the result is hard to reconcile with standard models in which expected returns rise with volatility, since it implies that the risk–return trade-off weakens precisely when volatility is high. They also reported that the market alpha remained positive after plausible transaction costs, with an estimated break-even cost of around 56 basis points, and that it survived a leverage cap of 1.5, although it fell to 3.10%.

2.5 Critiques of volatility management

The strongest critique is that the published alphas describe a combination that no investor could have held. Cederburg et al. (2020) point out that exploiting a positive spanning alpha requires combining the managed and unmanaged factors in proportions that depend on the full-sample regression coefficients, which are unknown in real time. Across 103 equity strategies, they find that volatility-managed portfolios do not systematically outperform their unmanaged counterparts in direct comparisons, and that realistic out-of-sample versions generally earn lower certainty-equivalent returns and Sharpe ratios than simple investments in the original portfolios. They attribute this poor out-of-sample performance primarily to structural instability in the underlying spanning regressions.

A second critique concerns trading costs. Volatility management implies large and frequent changes in exposure, and the factors themselves are costly long–short portfolios. Barroso and Detzel (2021) examine whether transaction costs, arbitrage risk and short-sale impediments explain the abnormal returns. They find that, even with six cost-mitigation strategies, volatility management of factors other than the market generally produces zero abnormal returns after costs and significantly reduces Sharpe ratios. The managed market portfolio is the exception: its abnormal returns are robust to costs and concentrated in periods of high investor sentiment.

The debate is not settled. DeMiguel et al. (2024) argue that the relevant investor holds several factors at once, and show that a conditional multifactor portfolio, which adjusts factor weights to market volatility, outperforms its unconditional counterpart even out of sample and after transaction costs. They also document that factor risk prices generally decrease with market volatility, which provides an economic rationale for volatility timing at the portfolio level rather than factor by factor.

2.6 Research gaps and positioning of the present study

The literature reviewed above leaves three gaps that the present study addresses. First, the existing critiques mostly re-analyse data that overlap with the original sample, so their tests are pseudo-out-of-sample: the researchers could observe the full history, including the periods on which the original result was established. Ten and a half years of new data are now available, from January 2016 to July 2026, that were not in the original sample and allow a genuinely out-of-sample test. Second, no study has evaluated the claim under a protocol fixed before the data

were examined, with explicit statistical power and decision rules. Without them, a non-significant out-of-sample result is easily over-interpreted as a refutation, and a significant result in one of many possible specifications is easily over-interpreted as a confirmation. Third, most studies report a small number of preferred specifications. The present study reports every one of its 288 pre-specified specifications, which allows the sensitivity of the conclusion to researcher choices, the fourth corollary of Ioannidis (2005), to be measured directly rather than discussed in the abstract.

3. Methodology

This chapter describes the research design of the study. It first presents the pre-registered hypotheses and the logic of the design, then the data sources and sample periods. It subsequently describes the construction of the volatility-managed strategies, including the real-time implementation and the treatment of trading costs, before setting out the statistical tests, the decision rules, the power analysis and the robustness grid.

3.1 Research design and pre-registration

The study was designed as a pre-registered replication and extension. Before any data were downloaded, a written plan fixed the research question, the hypotheses, the sample periods, the construction of the strategies, the statistical tests, the multiple-testing correction, the decision rules and the full robustness grid. The plan also included a deviation log, in which any later change had to be recorded with its date, its reason, and whether results had already been seen. Two deviations were recorded, both before any regression was estimated, and they are reported in Appendix B.

Four hypotheses were specified, as summarised in Table 1. H1 is a replication test: it requires the spanning alpha of the managed market factor, estimated on the original sample, to fall within two published standard errors of the published estimate. H2 asks whether the spanning alpha remains positive on the new data. H3 asks the more demanding question of whether an investor running the strategy in real time earns a higher Sharpe ratio than from the unmanaged factor. H4 asks whether H2 and H3 survive trading costs of 10 and 25 basis points per unit of turnover.

Table 1. Pre-registered hypotheses and pass conditions.

	Hypothesis	Period	Pass condition
H1	Replication: the managed market factor has a positive spanning alpha of the published size	1926–2015	α within two published SEs of 4.86% (1.74% to 7.98%)
H2	The spanning alpha remains positive on new data	2016–2026	$\alpha > 0$, one-sided $p < 0.05$, Holm–Bonferroni
H3	A real-time investor earns a higher Sharpe ratio from the managed factor	2016–2026 and 1936–2015 (Mom from 1937, RMW and CMA from 1973)	Sharpe gain > 0, bootstrap $p < 0.05$
H4	H2 and H3 survive trading costs	As above	Hold at 10 and 25 bps per unit of turnover

Note: The primary test is H3/H4 for momentum at 10 bps, evaluated at the 5% level without correction. All other tests are secondary. Source: Pre-registration plan dated 24 September 2026.

The primary test is H3 and H4 for the momentum factor at 10 basis points of trading costs, evaluated at the 5% level without correction. All other tests, on the remaining five factors, at other cost levels and for H2, are secondary and subject to a multiple-testing correction. The

choice of momentum as the primary factor was itself the product of the power analysis described in Section 3.6: it was originally the market factor, and the change was recorded before any data were obtained.

3.2 Data sources and sample periods

All data come from the Kenneth French Data Library (French, 2026), the same source used by Moreira and Muir (2017). Six files were used, as shown in Table 2: the daily and monthly versions of the Fama/French 3 Factors, the Momentum Factor, and the Fama/French 5 Factors (2x3). The market, size and value factors and the risk-free rate are taken from the three-factor file because it has the longest history, and the profitability and investment factors are taken from the five-factor file. All files carry the CRSP 202607 vintage, which means that the underlying stock database was updated through July 2026.

Table 2. Data files and sample periods.

File (Kenneth French Data Library)	Frequency	Series used	First observation
Fama/French 3 Factors	Monthly and daily	Mkt-RF, SMB, HML, RF	July 1926
Momentum Factor (Mom)	Monthly and daily	Mom	January 1927 (monthly), November 1926 (daily)
Fama/French 5 Factors (2x3)	Monthly and daily	RMW, CMA	July 1963

Note: All files carry the CRSP 202607 vintage and end in July 2026. They were downloaded on 24 September 2026 and archived with SHA-256 checksums. Source: French (2026).

Three periods are distinguished. The replication window reproduces the original sample and ends in April 2015. Although Moreira and Muir (2017) describe their sample as running from 1926 to 2015, their observation counts (1,065 months for the market, size and value factors, 1,060 for momentum and 621 for profitability and investment) are only consistent with a sample ending in April 2015, and this end date was adopted before any regression was run. The out-of-sample window runs from January 2016 to July 2026, 127 months that the original authors could not observe. The pre-2016 period, from the start of the data to December 2015, serves as the in-sample benchmark against which out-of-sample estimates are compared.

Two properties of the data deserve attention. First, the Kenneth French Data Library revises past factor returns whenever the underlying databases change. Since January 2025, returns have been computed from CRSP's new CIZ file format, in which monthly returns are compounded from daily returns, whereas the original paper used the legacy format. Small differences between our replication and the published figures are therefore expected even with identical code. To guarantee reproducibility, the downloaded files were archived with SHA-256 checksums, reported in Appendix D, and were never re-downloaded during the project. Second, the data were checked for completeness before use: no factor contains missing values after its

first observation, and the monthly market return in the file matches the compounded daily returns with a mean absolute difference of 0.05 percentage points.

3.3 Construction of volatility-managed portfolios

The construction follows Moreira and Muir (2017) exactly. For each factor f and each month t, the realised variance is computed from the daily factor returns within that month as the sum of squared deviations from their monthly mean:

$$\mathrm{RV}_t = \Sigma_{d \in t} (f_d - \bar{f}_t)^2$$

The volatility-managed return in the following month scales the factor's return by the inverse of this realised variance:

$$f^{\sigma}_{t+1} = (c / \mathrm{RV}_t) \times f_{t+1}$$

where c is a scaling constant and $w_t = c / \mathrm{RV}_t$ is the weight on the factor held during month t + 1. A weight of one corresponds to holding the unmanaged factor. The weight is set at the end of month t using only information available at that date, and it is applied to the return of month t + 1.

For the replication (H1), c is chosen, as in the original paper, so that the managed and unmanaged factors have the same volatility over the full sample. This choice uses information from the whole sample and could not have been made in real time, but it does not affect the t-statistic of the spanning alpha, which is why it is appropriate for replicating the published regressions. For the out-of-sample spanning regressions (H2), c is estimated once on all data before January 2016 and held fixed thereafter, so that no information from the test period enters the construction.

3.4 Real-time implementation and trading costs

The real-time strategy used for H3 and H4 re-estimates the scaling constant every month from an expanding window of past returns. The constant applied in month t + 1 is the ratio of the standard deviation of the unmanaged factor to that of the unscaled managed return, both computed from data up to and including month t. At least 120 months of history are required before the strategy starts, so the real-time series begins in August 1936 for the market, size and value factors, in January 1937 for momentum and in August 1973 for profitability and investment. This implementation corresponds to an investor who follows the rule from the start, learning the appropriate scale as data accumulate, and it is the real-time version emphasised by Cederburg et al. (2020).

Trading costs are charged on the change in exposure caused by the scaling. Turnover in month t + 1 is the absolute change in weight, $|w_{t+1} - w_t|$, and the net return equals the gross managed return minus the cost rate multiplied by turnover. Cost rates of 0, 10, 25 and 50 basis points per unit of turnover are considered. The unmanaged factor, with a constant weight of one, bears no scaling cost. The cost of rebalancing the factor portfolio itself is ignored for both

strategies. Because this cost is proportional to the size of the position, ignoring it favours the managed strategy in months when its weight exceeds one and the unmanaged factor in months when it is below one. Before 2016 the median weight exceeded one for the market, value and momentum, but after 2016 it lay between 0.25 and 0.67 for every factor (Table 10). The direction of the resulting bias therefore differs between periods and is ambiguous out of sample, and the net results should not be read as a bound in either direction.

3.5 Statistical tests and decision rules

Two families of tests are used. The spanning regressions of H1 and H2 regress the managed return on the unmanaged return:

$$f^{\sigma}_{t+1} = \alpha + \beta\, f_{t+1} + \varepsilon_{t+1}$$

The intercept α is annualised by multiplying by 12. Standard errors are heteroscedasticity-consistent (White, 1980), as in the original paper, and Newey–West standard errors with 12 lags (Newey & West, 1987) are computed as a check. For H2 the test is one-sided, with the null hypothesis that α is at most zero.

The Sharpe ratio comparisons of H3 and H4 test whether the difference between the annualised Sharpe ratios of the managed and unmanaged strategies is positive. Since monthly returns are serially dependent and fat-tailed, inference uses the stationary block bootstrap of Politis and Romano (1994), in the spirit of Ledoit and Wolf (2008). The managed and unmanaged return series are resampled jointly, preserving their contemporaneous correlation, in blocks of geometrically distributed length with a mean of 12 months, and 10,000 bootstrap samples are drawn with a fixed random seed (20260924) so that every number is exactly reproducible. The one-sided p-value is the share of centred bootstrap differences that exceed the observed difference, and the 95% confidence interval is the percentile interval of the bootstrap distribution. These two quantities are not fully consistent with each other. The centred p-value corresponds to a basic bootstrap interval rather than a percentile interval, and the two diverge when the bootstrap distribution is skewed. In addition, a two-sided 95% interval corresponds to a one-sided test at 2.5%, not 5%. The pre-registered p-value is retained for all decisions, but Appendix C also reports, as a post-hoc check, the one-sided p-value consistent with the percentile interval, which is the share of bootstrap differences at or below zero. For the primary test the two p-values are 0.45 and 0.44. As an independent check, the primary test is also computed with the asymptotic test of Jobson and Korkie (1981) in the corrected form of Memmel (2003).

The multiple-testing correction follows the logic of Harvey et al. (2016). The primary momentum test is evaluated alone at the 5% level. For each of the other hypothesis–cost–period combinations, the p-values of the five or six factors are adjusted with the sequentially rejective procedure of Holm (1979). In addition, every t-statistic above 3.0 is flagged.

Finally, every out-of-sample result is classified with one of three pre-registered labels. A result is *Supported* if it passes its test after correction. It is *Contradicted* if the 95% confidence interval of the out-of-sample estimate lies entirely below the corresponding pre-2016 estimate, which means the effect is significantly smaller on the new data. Otherwise it is *Inconclusive*. This third category is central to the design. With ten years of data, failing to find an effect is not evidence that the effect is absent, and the label prevents a low-powered null result from being reported as a refutation.

3.6 Power analysis

Before any data were obtained, the power of the out-of-sample test was estimated from the published results. The expected t-statistic of an alpha is approximately its annualised appraisal ratio multiplied by the square root of the number of years of data. Applied to the original sample, $0.34 \times \sqrt{90} \approx 3.2$ closely reproduces the published market t-statistic of 3.1, which supports the approximation. Table 3 applies it to the 10.5 years of new data.

Table 3. Statistical power of the out-of-sample test.

Scenario	Effect size	Expected t (10.5 years)	Power, one-sided 5%	Years for 80% power
A. Spanning α (H2), pre-registered				
Market, effect as published	AR 0.34	1.10	29%	53
Market, effect shrinks by 58%	AR 0.14	0.46	12%	303
Momentum, effect as published	AR 0.875	2.84	88%	8
B. Computed after the results were known (post hoc)				
Momentum spanning α, calibrated to the published t of 7.32	AR 0.78	2.51	81%	10
Momentum real-time Sharpe gain at 10 bps (primary test), true gain +0.13	ΔSR 0.13	0.71	18%	130

Note: Panel A: expected t ≈ appraisal ratio (AR) × √years, with published appraisal ratios from Moreira and Muir (2017); computed before any data were downloaded. The 58% shrinkage is the average post-publication decline in McLean and Pontiff (2016). Panel B was computed after the results were known. The first row scales the published momentum t-statistic (12.51 / 1.71) by √(10.5 / 89), because AR × √years overstates it (8.2 against 7.3). The second row uses the standard error of the primary test implied by its bootstrap confidence interval (0.18) and, as the true effect, the gain of the same real-time strategy before 2016 (+0.13). Source: Author's own calculations.

The results changed the design. If the market effect is exactly as large as published, the out-of-sample test has only a 29% chance of detecting it. If the effect has decayed by the 58% average reported by McLean and Pontiff (2016), the probability falls to 12%, and more than three centuries of data would be required for conventional power. The momentum effect, with an appraisal ratio of 0.875, can be detected with 88% probability. The primary test was therefore moved from the market factor to momentum, the factor for which, according to this analysis, the

new data could most plausibly settle the question. This decision was taken before any data were downloaded and is recorded in Appendix B.

Two qualifications became apparent only after the results were known, and Panel B of Table 3 reports them. First, the approximation overstates the in-sample momentum t-statistic ($0.875 \times \sqrt{89} \approx 8.2$, against a published 7.3). Calibrated to the published t-statistic, the power of the momentum spanning test is 81% rather than 88%. Second, and more importantly, the power analysis concerns the spanning alpha of H2, not the real-time Sharpe ratio gain that became the primary test. The Sharpe ratio gain is a different statistic with a much smaller expected effect. The confidence interval of the primary test implies a standard error of about 0.18, so a true gain of +0.13, the gain of the same real-time strategy before 2016, would be detected only about 18% of the time, and about 130 years of data would be needed for 80% power. The move to momentum raised the power of the spanning test, but the primary test remained weak. Its *Inconclusive* result in Section 4.4 is the outcome such a test would most likely produce whether or not the effect persists.

3.7 Robustness grid

The specification of a volatility-managed strategy involves several choices that the original paper fixed but that a researcher could reasonably vary: the length of the window used to measure variance, whether to scale by variance or by volatility, whether to cap leverage, and which cost level to assume. Instead of reporting a few selected alternatives, the study pre-specified a full grid of 48 combinations per factor, shown in Table 4, giving 288 specifications in total. Every specification is estimated as a real-time strategy on the out-of-sample window, and every result is reported. Longer variance windows sum squared deviations over the last three or six months of daily returns, the volatility scaling uses the square root of realised variance, and the leverage cap limits the weight to 1.5.

Table 4. Pre-registered robustness grid.

Choice	Options (primary specification in bold)
Variance window	**1 month**, 3 months, 6 months of daily returns
Scaling	**1 / variance**, 1 / volatility
Leverage	**Uncapped**, capped at 1.5
Trading cost per unit of turnover	0, **10**, 25, 50 bps

Note: 3 × 2 × 2 × 4 = 48 specifications per factor, 288 in total, all reported. Source: Pre-registration plan.

The grid serves two purposes. The first is conventional: a result that holds only for the primary specification is fragile. The second follows directly from Ioannidis (2005): the grid measures how many "significant" results a researcher could have obtained by choosing among equally defensible specifications after seeing the data. It was agreed in advance that no option would be added to the grid after results were seen, and that a result holding only in some specifications would be reported as fragile, however significant it appeared there.

All computations were performed in Python using pandas, NumPy and statsmodels (Seabold & Perktold, 2010), and figures were produced with Matplotlib (Hunter, 2007). The complete code, which reproduces every table and figure in under one minute, is provided with the paper.

4. Results

This chapter presents the empirical findings in five sections. Section 4.1 provides descriptive statistics for the factors before and after 2016. Section 4.2 reports the replication of the original results. Section 4.3 presents the out-of-sample spanning regressions, and Section 4.4 the primary real-time Sharpe ratio tests. Section 4.5 reports the full robustness grid.

4.1 Descriptive statistics

Table 5 presents summary statistics of the monthly returns of the six factors, separately for the period before 2016 and for the out-of-sample window.

Table 5. Descriptive statistics of monthly factor returns, before and after 2016.

Factor	Period	Months	N	Mean (%/yr)	SD (%/yr)	Sharpe	Min (%/mo)	Max (%/mo)	Skew
Mkt	Pre-2016	1926-07 to 2015-12	1074	7.78	18.66	0.42	−28.74	38.81	0.20
	2016–2026	2016-01 to 2026-07	127	12.98	15.74	0.82	−13.37	13.60	−0.36
SMB	Pre-2016	1926-07 to 2015-12	1074	2.42	11.06	0.22	−17.39	35.96	1.87
	2016–2026	2016-01 to 2026-07	127	−1.69	9.44	−0.18	−5.93	7.14	0.40
HML	Pre-2016	1926-07 to 2015-12	1074	4.69	12.20	0.38	−13.19	35.52	2.33
	2016–2026	2016-01 to 2026-07	127	0.66	13.17	0.05	−13.83	12.86	0.11
Mom	Pre-2016	1927-01 to 2015-12	1068	8.13	16.51	0.49	−52.61	18.05	−3.19
	2016–2026	2016-01 to 2026-07	127	1.38	13.97	0.10	−16.21	9.51	−0.84
RMW	Pre-2016	1963-07 to 2015-12	630	3.12	7.71	0.40	−18.93	13.04	−0.45
	2016–2026	2016-01 to 2026-07	127	2.90	8.84	0.33	−8.16	11.05	0.50
CMA	Pre-2016	1963-07 to 2015-12	630	3.54	6.85	0.52	−6.63	9.00	0.29
	2016–2026	2016-01 to 2026-07	127	0.07	8.60	0.01	−7.06	7.73	0.29

Note: Returns are monthly factor returns from the Kenneth French Data Library (CRSP 202607 vintage). Means are multiplied by 12 and standard deviations by √12. The Sharpe ratio is annualised; factor returns are already excess or zero-cost returns. Source: Author's own calculations.

The most striking feature of Table 5 is how differently the factors behaved after 2016. The market factor had an exceptional decade, with an average excess return of 12.98% per year and a Sharpe ratio of 0.82, almost twice its long-run value of 0.42. Most of the other factors, in contrast, performed poorly. The size factor had a negative average return of −1.69% per year, and the value and investment factors earned close to zero, with Sharpe ratios of 0.05 and 0.01. Momentum, the focus of the primary test, fell from an average of 8.13% per year and a Sharpe ratio of 0.49 before 2016 to 1.38% and 0.10 afterwards. Only the profitability factor kept a performance close to its historical level, with a Sharpe ratio of 0.33 against 0.40.

The volatility of the factors also changed. The median realised volatility of momentum, measured from daily returns within each month, roughly doubled from 6.97% to 13.60% per year, and similar increases occurred for size, value, profitability and investment. This matters for the managed strategy, whose weight is inversely proportional to recent variance: in a regime

of persistently higher volatility, a strategy calibrated on history holds less of the factor. The extreme months of the early sample, such as the −52.61% momentum return of August 1932 and the 38.81% market return of April 1933, have no equivalent after 2016, where the largest monthly momentum loss is −16.21%.

4.2 Replication of the original results

Table 6 compares the spanning regressions estimated in this study on the replication window with those published in Table I, Panel A, of Moreira and Muir (2017).

Table 6. Replication of Moreira and Muir (2017), Table I, Panel A (H1).

	N		α (%/yr)		SE		β		R^2		Gap
Factor	**Ours**	**Paper**	**Ours**	**Paper**	**Ours**	**Paper**	**Ours**	**Paper**	**Ours**	**Paper**	**(SEs)**
Market (Mkt)	1,065	1,065	4.93	4.86	1.57	1.56	0.60	0.61	0.36	0.37	+0.05
Size (SMB)	1,065	1,065	−0.59	−0.58	0.91	0.91	0.60	0.62	0.37	0.38	−0.01
Value (HML)	1,065	1,065	1.73	1.97	1.05	1.02	0.54	0.57	0.29	0.32	−0.24
Momentum (Mom)	1,060	1,060	12.83	12.51	1.72	1.71	0.45	0.47	0.20	0.22	+0.19
Profitability (RMW)	621	621	2.80	2.44	0.89	0.83	0.58	0.62	0.34	0.38	+0.44
Investment (CMA)	621	621	0.36	0.38	0.67	0.67	0.67	0.68	0.45	0.46	−0.03

Note: Spanning regressions of the volatility-managed factor on the unmanaged factor, $f^{\sigma} = \alpha + \beta f + \varepsilon$, with c chosen so that both have the same full-sample volatility. Samples end in April 2015 and start in August 1926 (Mkt, SMB, HML), January 1927 (Mom) and August 1963 (RMW, CMA). Standard errors are heteroscedasticity-consistent (White, 1980). Gap = (our α − published α) / published SE. Source: Author's own calculations.

The replication succeeds almost perfectly, and hypothesis H1 is supported. The number of observations matches the published counts exactly for all six factors, which confirms the April 2015 end date. The market alpha is 4.93% per year against 4.86% published, with a standard error of 1.57 against 1.56, and falls well within the pre-registered interval of 1.74% to 7.98%. The momentum alpha is 12.83% against 12.51%, with a t-statistic of 7.47. Every alpha lies within half a published standard error of the original, and the betas and R^2 values differ by at most 0.04. The remaining small differences are consistent with the revisions to the underlying CRSP data described in Section 3.2.

The replication also reproduces a feature of the original results that is easily overlooked. The managed portfolio does not have a higher Sharpe ratio than the unmanaged factor for every factor. For size, value and investment, the full-sample Sharpe ratios of the managed versions (0.08, 0.35 and 0.41) are lower than those of the unmanaged factors (0.23, 0.39 and 0.54), even though the value factor has a positive alpha. A positive spanning alpha indicates that the managed portfolio is useful in combination with the unmanaged one, not that it is better on its own. This distinction motivates the Sharpe ratio tests of H3.

4.3 Out-of-sample spanning regressions

Table 7 reports the spanning alphas on the out-of-sample window, with the scaling constant fixed on pre-2016 data, together with the pre-registered labels. The full regression output is reported in Appendix A.

Table 7. Out-of-sample spanning alphas (H2 and H4).

	Gross				10 bps		25 bps	
Factor	**α (pre-2016)**	**95% CI**	**p**	**Label**	**α (pre-2016)**	**Label**	**α (pre-2016)**	**Label**
Mkt	1.48 (4.88)	[−7.45, 10.40]	0.373	Inconclusive	0.63 (3.97)	Inconclusive	−0.65 (2.62)	Inconclusive
SMB	−0.71 (−0.58)	[−2.34, 0.91]	0.806	Inconclusive	−0.93 (−1.24)	Inconclusive	−1.26 (−2.23)	Inconclusive
HML	−1.11 (1.65)	[−2.88, 0.66]	0.890	Contradicted	−1.32 (0.83)	Contradicted	−1.63 (−0.41)	Inconclusive
Mom	0.51 (12.80)	[−1.51, 2.54]	0.310	Contradicted	0.25 (11.67)	Contradicted	−0.14 (9.98)	Contradicted
RMW	0.36 (2.81)	[−0.63, 1.35]	0.238	Contradicted	0.16 (2.06)	Contradicted	−0.13 (0.94)	Contradicted
CMA	−0.81 (0.28)	[−2.45, 0.83]	0.833	Inconclusive	−1.10 (−0.40)	Inconclusive	−1.54 (−1.43)	Inconclusive

Note: α in % per year. The scaling constant c is estimated once on all data before January 2016 and held fixed. Out-of-sample window: January 2016 to July 2026 (127 months). p is one-sided (H0: α ≤ 0) with White standard errors; Holm–Bonferroni correction across the six factors. Labels follow the pre-registered rules: Supported = significant after correction; Contradicted = 95% CI entirely below the pre-2016 estimate; otherwise Inconclusive. Each α is the 2016–2026 estimate; the pre-2016 α at the same cost level, which is the benchmark for the Contradicted label, is shown in parentheses. The 95% CI and p refer to the gross 2016–2026 α. Source: Author's own calculations.

Hypothesis H2 is not supported for any factor. No out-of-sample alpha is significantly positive, before or after trading costs. More importantly, three results are classified as *Contradicted*. The alpha of managed momentum falls from 12.80% per year before 2016 to 0.51% afterwards, with a 95% confidence interval from −1.51% to 2.54% that lies entirely below the in-sample estimate. The profitability alpha falls from 2.81% to 0.36%, and the value alpha turns negative, from 1.65% to −1.11%. Under the pre-registered rule, the new data for these factors are inconsistent with an alpha of the published size, measured in percent per year.

This comparison, however, mixes two effects. The scaling constant is fixed on pre-2016 data, and the factors were more volatile relative to that constant after 2016, so the managed portfolios held much smaller positions (median weights between 0.24 and 0.67 with the constant fixed on pre-2016 data, Table 8) and mechanically earned smaller alphas in percent per year. A risk-adjusted comparison, which was not pre-registered, asks what the out-of-sample alpha would have been if the pre-2016 appraisal ratio, the alpha per unit of residual risk, had persisted. Table 8 reports it. For momentum, the implied alpha of 2.90% lies above the upper bound of the confidence interval (2.54%), so the decline survives the adjustment, although narrowly: the out-of-sample appraisal ratio is 0.15, against 0.87 before 2016. For profitability (an implied 0.80% against an upper bound of 1.35%) and value (0.48% against 0.66%), it does not. The labels in Table 7 follow the pre-registered rule and are kept, but only the momentum result should be read as evidence that the efficiency of the strategy declined. For profitability and value, the lower alphas are also consistent with smaller positions at unchanged efficiency.

Table 8. Out-of-sample alphas against a risk-adjusted benchmark (post-hoc analysis).

Factor	Median weight, 2016–2026	Appraisal ratio, pre-2016	Appraisal ratio, 2016–2026	Residual volatility, 2016–2026	α implied by pre-2016 appraisal ratio	Upper 95% bound of 2016–2026 α	Below benchmark?
Mkt	0.67	0.33	0.12	12.67	4.14	10.40	No
SMB	0.31	−0.07	−0.26	2.74	−0.18	0.91	No
HML	0.27	0.16	−0.37	2.99	0.48	0.66	No
Mom	0.24	0.87	0.15	3.34	2.90	2.54	Yes
RMW	0.26	0.45	0.20	1.78	0.80	1.35	No
CMA	0.33	0.06	−0.29	2.74	0.15	0.83	No

Note: Appraisal ratio = spanning α divided by residual volatility (both in % per year, gross of costs). Median weight = median position in the factor of the managed strategy with c fixed on pre-2016 data, as in Table 7. The implied α is the α the managed factor would have earned from January 2016 to July 2026 if its pre-2016 appraisal ratio had persisted, given its out-of-sample residual volatility. This comparison was not pre-registered. Source: Author's own calculations.

One label in Table 7 also requires explanation. Value is *Contradicted* at 10 basis points (α = −1.32%) but *Inconclusive* at 25 basis points (α = −1.63%), even though the alpha is lower. The reason is that the benchmark moves as well: after costs, the pre-2016 alpha of managed value falls from 0.83% at 10 basis points to −0.41% at 25 basis points, which lies inside the out-of-sample confidence interval (upper bound 0.16%). Table 7 therefore reports the after-cost pre-2016 benchmark in parentheses next to each after-cost alpha.

The market alpha illustrates the opposite situation. Its point estimate of 1.48% is positive, but its confidence interval runs from −7.45% to 10.40%. The interval includes both zero and the in-sample estimate of 4.88%, and the result is therefore *Inconclusive*. This width is not an accident of the data: it is the low power predicted in Section 3.6. The out-of-sample market test was never able to distinguish between a strategy that still works and one that does not, which is precisely why it was not chosen as the primary test.

4.4 Real-time Sharpe ratio tests

Table 9 reports the primary results of the study: the difference in annualised Sharpe ratio between the real-time managed strategy and the unmanaged factor over the out-of-sample window.

Table 9. Real-time Sharpe ratio comparison, January 2016 to July 2026 (H3 and H4).

	Sharpe, 10 bps		Gain	Gain at 10 bps					Gain
Factor	Plain	Managed	gross	Estimate	95% CI	p	Pre-2016	Label	25 bps
Mkt	0.82	0.54	−0.23	−0.29	[−0.83, 0.37]	0.839	−0.07	Inconclusive	−0.37
SMB	−0.18	−0.35	−0.12	−0.17	[−0.58, 0.16]	0.802	−0.14	Inconclusive	−0.24
HML	0.05	−0.24	−0.25	−0.29	[−0.64, 0.05]	0.949	−0.16	Inconclusive	−0.35
Mom †	0.10	0.13	+0.08	+0.03	[−0.32, 0.39]	0.447	+0.13	Inconclusive	−0.06
RMW	0.33	0.30	+0.05	−0.03	[−0.40, 0.29]	0.574	+0.13	Inconclusive	−0.14
CMA	0.01	−0.27	−0.21	−0.28	[−0.59, 0.09]	0.964	−0.18	Inconclusive	−0.39

Note: Real-time strategy: c re-estimated each month on an expanding window, 120-month burn-in. Gain = managed minus unmanaged annualised Sharpe ratio. p is one-sided (H0: gain ≤ 0) from a stationary block bootstrap (10,000 resamples, mean block length 12 months, seed 20260924). † Primary test, evaluated at 5% without correction; the other five factors use Holm–Bonferroni. Pre-2016 = the same real-time strategy from the end of the burn-in to December 2015. At 25 bps, the 95% CI lies entirely below zero only for HML ([−0.70, −0.01]; pre-2016 gain −0.26) and CMA ([−0.70, −0.03]; pre-2016 gain −0.35); at every cost level, every interval contains the pre-2016 gain. Source: Author's own calculations.

The primary pre-registered test is *Inconclusive*. After 10 basis points of trading costs, managed momentum earned a Sharpe ratio of 0.13 against 0.10 for plain momentum, a gain of +0.03. The 95% bootstrap confidence interval runs from −0.32 to +0.39, and the one-sided p-value is 0.45. The asymptotic test of Jobson and Korkie (1981) with the correction of Memmel (2003) gives an almost identical p-value of 0.45. Hypotheses H3 and H4 are therefore not supported for momentum: the data provide no evidence that the managed strategy outperforms, but they are also unable to rule out a gain of the size achieved by the same real-time strategy before 2016 (+0.13).

For the other five factors, the results are uniformly unfavourable. The point estimates are negative for the market (−0.29), size (−0.17), value (−0.29), investment (−0.28) and, after costs, profitability (−0.03). None is labelled *Contradicted*. That label does not depend on statistical significance: it requires the confidence interval to lie entirely below the pre-2016 estimate, and for every factor and cost level the interval contains the pre-2016 gain of the same real-time strategy. The negative point estimates are therefore not evidence that the managed strategies deteriorated relative to the same strategies before 2016. For value and investment at 25 basis points, the interval lies just below zero (upper bounds −0.01 and −0.03, note to Table 9), which suggests that the managed strategy earned a lower Sharpe ratio than the plain factor after 2016. This comparison is secondary and was not pre-registered, and the intervals are uncorrected; with a Holm–Bonferroni correction across the six factors, results this close to zero would very likely not survive. Both gains were already negative before 2016 (−0.26 and −0.35), so this is not a change relative to the original period. In no case, however, does the managed strategy outperform. For the market factor in particular, the managed strategy earned a Sharpe ratio of 0.54 against 0.82 for simply holding the market. Its reduced exposure during the volatile but profitable episodes of the decade was costly.

A second result emerges from the pre-2016 column of Table 9, which concerns the historical period and not the new data. Even before 2016, the real-time strategy delivered a much smaller advantage than the full-sample results of Table 6 suggest. For momentum, the full-sample managed strategy of Section 4.2 improved the Sharpe ratio by 0.51. When the scaling constant must be learned in real time, the gross improvement over 1937–2015 falls to 0.19, and to 0.13 after 10 basis points of costs. Starting the comparison in 1937 instead of 1927 explains little of the drop: with the full-sample constant, the improvement over 1937–2015 is still 0.45. The loss comes from the real-time estimation itself. In the early decades, the expanding-window estimate of the constant was two to five times larger than its long-run value,

and the strategy took extreme positions, with weights reaching almost 30 times the factor in calm months. Table 10 documents these weights.

Table 10. Real-time portfolio weights and turnover.

Factor	Period	Median w	99th pct. w	Max w	Mean monthly turnover	Cost at 10 bps (%/yr)
Mkt	Pre-2016	1.12	8.54	18.93	0.97	1.16
	2016–2026	0.67	3.78	4.25	0.69	0.83
SMB	Pre-2016	0.87	6.71	12.65	0.76	0.91
	2016–2026	0.32	1.19	1.24	0.18	0.22
HML	Pre-2016	1.66	9.34	14.39	1.15	1.37
	2016–2026	0.28	1.25	1.55	0.18	0.21
Mom	Pre-2016	1.37	12.91	29.58	1.43	1.72
	2016–2026	0.25	1.43	1.69	0.22	0.26
RMW	Pre-2016	0.69	3.01	5.35	0.39	0.47
	2016–2026	0.28	1.26	1.45	0.17	0.20
CMA	Pre-2016	0.83	4.77	7.84	0.52	0.63
	2016–2026	0.35	1.58	2.60	0.25	0.30

Note: w is the weight on the factor held in a month (1 = the unmanaged factor). Turnover is $|w_t - w_{t-1}|$. Pre-2016 starts after the 120-month burn-in. Source: Author's own calculations.

After 2016, the picture is reversed. The real-time weights on momentum were small, with a median of 0.25 and a maximum of 1.69, because the strategy had learned a scale from a history of calmer periods, while realised volatility after 2016 was higher. This is also why the cost drag was modest in the out-of-sample period, at 0.26% per year at 10 basis points for momentum. Trading costs are not what keeps the gain small after 2016: the gross gain is already only +0.08.

Figure 1 places the out-of-sample period in historical context by plotting the Sharpe ratio gain of the real-time strategy after 10 basis points of costs over rolling ten-year windows.

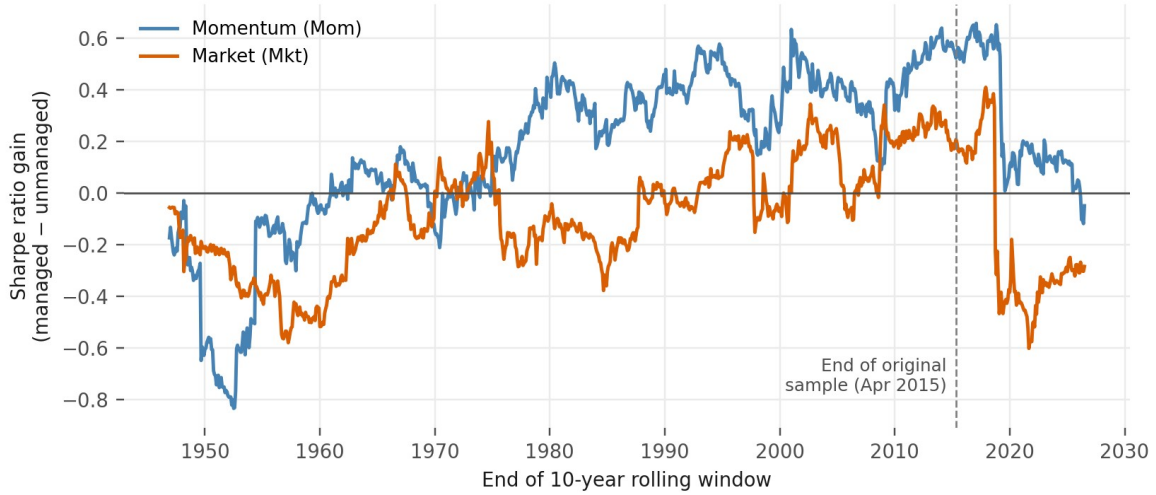


Figure 1. Sharpe ratio gain of the real-time volatility-managed strategy over rolling 10-year windows, after 10 bps of trading costs. Source: Author's own calculations. Figure produced using Matplotlib (Hunter, 2007).

The gain of managed momentum was negative in the early decades, rose steadily from the 1960s, and peaked at 0.66 for the window ending in February 2017, shortly after the end of the original sample. It then fell sharply in April 2019, when April 2009 left the ten-year window. In that month momentum lost 34.4%, while the managed strategy, holding only 1.5% of the factor after the turbulence of the preceding months, lost 0.5%. It has since drifted towards zero. The market factor shows a similar but weaker pattern. The figure suggests that much of the historical advantage of volatility management came from a small number of crash episodes, and that the original paper was published at the historical peak of the strategy's performance.

4.5 Robustness and specification search

Table 11 summarises the out-of-sample Sharpe ratio gains across the 288 specifications of the pre-registered robustness grid, and Figure 2 displays every specification individually.

Table 11. Summary of the pre-registered robustness grid (out-of-sample Sharpe ratio gain).

Factor	Specs	Gain > 0	p < 0.05	Median gain	Range	Median pre-2016 gain
Market (Mkt)	48	0%	0	−0.16	−0.50 to 0.00	−0.02
Size (SMB)	48	0%	0	−0.08	−0.35 to −0.03	−0.07
Value (HML)	48	0%	0	−0.23	−0.46 to −0.11	0.00
Momentum (Mom)	48	88%	26	+0.16	−0.19 to +0.29	+0.25
Profitability (RMW)	48	8%	0	−0.07	−0.33 to +0.05	+0.11
Investment (CMA)	48	0%	0	−0.25	−0.59 to −0.12	−0.04
All factors	**288**	**16%**	**26**	**−0.13**	**−0.59 to +0.29**	**0.00**

Note: 48 specifications per factor = 3 variance windows (1, 3, 6 months) × 2 scalings (1/variance, 1/volatility) × 2 leverage rules (uncapped, cap at 1.5) × 4 trading costs (0, 10, 25, 50 bps). All are real-time strategies. p < 0.05 counts uncorrected one-sided bootstrap p-values. Source: Author's own calculations.

For five of the six factors, the robustness grid confirms the main results. For the market, size, value and investment factors, not one of the 192 specifications produces a positive out-of-sample gain, and only 4 of the 48 profitability specifications do so. The conclusion that volatility management did not help these factors after 2016 is therefore not an artefact of the primary specification.

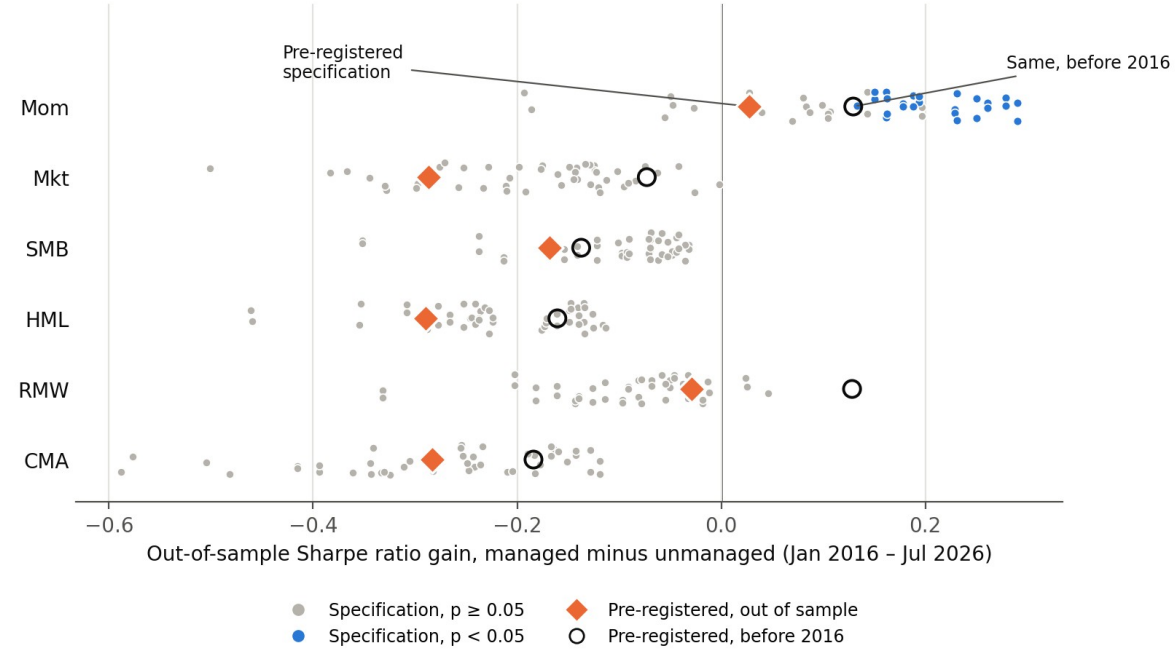


Figure 2. Out-of-sample Sharpe ratio gain of all 288 real-time specifications, January 2016 to July 2026. Source: Author's own calculations. Figure produced using Matplotlib (Hunter, 2007).

Momentum is different, and it provides the most instructive result of the study. Across its 48 specifications, 88% produce a positive gain, and 26 are "significant" with an uncorrected one-sided p-value below 0.05. This count overstates the evidence in three ways. First, the specifications are far from independent: trading costs shift the gain almost deterministically, and for three- and six-month windows the leverage cap never binds, so capped and uncapped results are identical. The 26 therefore amount to four distinct strategies, the three- and six-month windows with variance or volatility scaling. Second, the pre-registered p-value is not fully consistent with the percentile confidence interval (Section 3.5). By the percentile-consistent p-value, 20 specifications, or three distinct strategies, fall below 0.05. Third, only 10 have a two-sided 95% confidence interval that excludes zero. All 26 use a three- or six-month variance window. None of the 16 specifications with the pre-registered one-month window reaches significance: their largest gain is +0.11, with a p-value of 0.10. The best specification, a six-month window with variance scaling, gains +0.28 at 10 basis points, with a confidence interval from 0.01 to 0.54 and a p-value of 0.019. It is identical with and without the leverage cap. The full set of momentum specifications is reported in Appendix C.

A researcher free to choose the variance window after seeing the data could therefore have reported that volatility-managed momentum "significantly outperforms out of sample, even after costs". The result would have been reproducible, and every individual step of the analysis would have been defensible. Yet it would have been produced by selecting one of 48 correlated specifications, and it would not have been a test of the pre-registered hypothesis. For the same reason, the result cannot be treated as a finding of this study. It is reported as exploratory, and as a hypothesis for future out-of-sample tests.

There is nevertheless one point in favour of longer windows that should be stated fairly. They also performed better before 2016: the median pre-2016 real-time gain of momentum is 0.19 with a one-month window, 0.26 with three months and 0.29 with six months. Longer windows also imply much lower turnover (a median of 0.05 per month for six months against 0.22 for one month). A study designed today might reasonably choose a longer window on these grounds. The point is not that longer windows are wrong, but that the choice had to be made before seeing the out-of-sample data for the result to count as evidence.

5. Discussion

This chapter interprets the results of Chapter 4. It first summarises the key findings, then discusses why the effect weakened after publication, what the study reveals about false findings in the sense of Ioannidis (2005), and finally the limitations of the study.

5.1 Key findings and interpretation

The study produced four main findings. First, the original results of Moreira and Muir (2017) are fully reproducible: with free public data and a few hundred lines of code, every published alpha was matched to within half a standard error. This is an important and positive result. The original finding was not the product of coding errors or of a proprietary dataset. Second, the spanning alphas that made the result famous largely disappeared on new data. By the pre-registered rule, the out-of-sample alphas of momentum, profitability and value are significantly smaller than their in-sample counterparts, but only for momentum does the decline survive, narrowly, a post-hoc adjustment for the smaller positions the managed portfolios held after 2016. Third, the question that matters to an investor, whether the managed strategy delivers a higher Sharpe ratio when run in real time, receives no positive answer for any factor. Every Sharpe ratio result is *Inconclusive*. The point estimates are negative for five factors and close to zero for momentum, but the tests, including the primary test with a power of only about 18%, cannot distinguish a persisting gain from its absence. Fourth, the conclusion for momentum depends on a choice, the variance window, which a less constrained analysis could have made after seeing the results.

Taken together, these findings are consistent with the critiques of Cederburg et al. (2020) and Barroso and Detzel (2021), and extend them to a period that neither study could observe. The gap between the full-sample and real-time results before 2016 (a momentum Sharpe ratio gain of 0.51 against 0.19) confirms, on the original data, that the spanning regression overstates what an investor could have achieved. The out-of-sample results provide no evidence that even this reduced real-time advantage persisted, although, with a power of about 18%, they cannot rule out that it did.

5.2 Why did the effect weaken?

Three explanations are compatible with the evidence, and they are not mutually exclusive. The first is statistical: part of the in-sample performance was luck. McLean and Pontiff (2016) estimate that anomaly returns fall by about a quarter after the end of the original sample for purely statistical reasons, and published estimates are selected for being large. The rolling-window evidence of Figure 1, in which the momentum gain peaked just as the original sample ended, is consistent with this selection effect.

The second explanation is a change in the market environment. Volatility management pays off when high volatility predicts low subsequent returns, as in the momentum crashes of 1932 and 2009 documented by Daniel and Moskowitz (2016). The out-of-sample period

contained no crash of comparable magnitude for momentum, while volatility was persistently higher than in the calibration period. In such an environment, a volatility-managed strategy holds a reduced position throughout, pays the cost of adjustment, and never collects the insurance payoff for which it was designed. This explanation implies that the strategy may still be valuable, but that its value is concentrated in rare events, and a decade is too short to observe them reliably.

The third explanation is that the effect was arbitraged away after publication. Volatility targeting has become widespread in the asset management industry, and McLean and Pontiff (2016) attribute the additional post-publication decline in anomaly returns to investors trading on academic research. The present study cannot distinguish this mechanism from the other two, since all three predict weaker performance after 2016. The fact that the replication succeeds while the out-of-sample alpha declines is consistent with each of them.

5.3 Lessons in light of Ioannidis (2005)

The study offers a concrete illustration of each of the corollaries of Ioannidis (2005). The most important lesson concerns statistical power. Table 12 applies his formula for the positive predictive value to the two power levels encountered in this study.

Table 12. Positive predictive value of a significant finding under Ioannidis (2005).

Prior odds R	Power (1 − β)	PPV, no bias (u = 0)	PPV, moderate bias (u = 0.3)
1:10	18%	0.26	0.11
1:10	29%	0.37	0.13
1:10	81%	0.62	0.21
1:4	18%	0.47	0.24
1:4	29%	0.59	0.27
1:4	81%	0.80	0.39

Note: α = 0.05. The power levels are those of this study's primary test, the real-time momentum Sharpe gain (18%), of the market spanning test (29%) and of the momentum spanning test calibrated to the published t-statistic (81%). Prior odds of 1:10 mean that one relationship in eleven is true. Formula from Ioannidis (2005), Table 2. Source: Author's own calculations.

With the power of the market-factor test (29%) and prior odds of one to ten, a significant result would have been true with a probability of only 37%, and with a moderate degree of bias, only 13%. At the 18% power of the primary momentum test, the corresponding values are 26% and 11%. Even at the 81% power of the momentum spanning test, bias reduces the PPV from 0.62 to 0.21. The practical implication is that a single significant out-of-sample result, especially in a test with low power, would have been weak evidence that the strategy works. Symmetrically, a non-significant result in a low-powered test is weak evidence that it does not. The three-way labelling rule adopted here, with its explicit *Inconclusive* category, is a direct response to this problem, and it is the reason why the market-factor results are not reported as a refutation.

The robustness grid illustrates the fourth corollary, on flexibility in design. Twenty-six of the 288 specifications, or 9%, produced a significant out-of-sample gain by the pre-registered p-value, all for momentum, although they reduce to four distinct strategies and only 10 have a 95% confidence interval that excludes zero. In isolation, any of them could have been presented as a successful out-of-sample confirmation. The pre-registration plan is what distinguishes the primary test from these alternatives, and the deviation log is what allows the reader to check that the plan was followed, subject to the caveat on its timing discussed in Section 5.4. The value of pre-registration lies less in any single result than in making the number of paths visible, as Simmons et al. (2011) and Nosek et al. (2018) argue.

Finally, the study shows that a failure to generalise is not the same as a false finding. Moreira and Muir (2017) reported a correct result: the in-sample spanning alphas exist in the data and are reproduced here. What does not hold is the inference, natural for a reader, that an investor could have captured them, or that they would continue. This distinction matters for the replication debate. The optimistic conclusion of Jensen et al. (2023), that most factors replicate, and the pessimistic evidence of Hou et al. (2020) and McLean and Pontiff (2016), that returns shrink or disappear, can both be true, because they answer different questions. This study finds both at once: the result replicates perfectly, but the momentum spanning alpha declines out of sample and no real-time gain is detectable.

5.4 Limitations

The study has several limitations. The most fundamental is the length of the out-of-sample window. With 127 months, even the momentum spanning test has good power only for an alpha of the published size, and the primary test, the real-time Sharpe ratio gain, has a power of only about 18% against a gain of the size achieved before 2016 (Table 3, Panel B). A smaller but economically meaningful effect could exist and remain undetected, and the confidence interval of the primary test (−0.32 to +0.39) includes both. The results should be revisited as new data accumulate, ideally with the same pre-registered protocol.

Second, the treatment of trading costs is simplified. Costs are charged only on the change in exposure caused by the scaling, at a fixed rate per unit of turnover, and the cost of trading the long–short factor portfolios themselves is ignored. The direction of the resulting bias is not clear. More realistic models of scaling costs, such as those of Barroso and Detzel (2021), would lower the net performance of the managed strategies. The omitted cost of trading the factor itself, however, is proportional to the position, and after 2016 the managed strategies held much less of each factor than the unmanaged benchmark, so including it would weigh more heavily on the benchmark. The net effect on the comparisons is therefore ambiguous in sign.

Third, the study examines each factor separately, as Moreira and Muir (2017) did. DeMiguel et al. (2024) argue that the benefits of volatility management appear more clearly at the level of a multifactor portfolio, and the present results do not test that claim. Fourth, the real-time implementation used here, an expanding-window estimate of the scaling constant, is one of

several possible choices, and the early-sample instability documented in Section 4.4 suggests that alternatives, such as a fixed volatility target, might behave differently in real time. Finally, the data themselves are revised over time, and the switch of the underlying database to the CIZ format in 2025 means that the historical returns used here differ slightly from those available to the original authors. The close match of the replication suggests that this effect is small.

A last limitation concerns the pre-registration itself. The plan, the download of the data, the two recorded deviations and the analysis are all dated 24 September 2026, and the plan was not lodged with a public registry such as OSF or AsPredicted. The reader must therefore take the order of events on trust: the checksums in Appendix D fix the data, but not the time at which the plan was written. Moreover, the broad performance of equity factors after 2016 was public knowledge before the plan was written, so the author was not blind to the outcome even before the data were downloaded. The design choices were taken from the original paper wherever possible, which limits the scope for such knowledge to shape them, but it cannot be excluded. Future out-of-sample tests of this kind should be registered publicly before the analysis begins.

6. Conclusion

This study set out to test one of the best-known results in recent empirical finance under conditions designed to protect the conclusion from the problems identified by Ioannidis (2005). It replicated the volatility-managed portfolios of Moreira and Muir (2017) from free public data and then tested them on the ten and a half years of returns that the original authors could not observe, under a protocol, a power analysis and a set of decision rules fixed before any data were downloaded.

The answer to the question in the title is largely negative. The published results replicate almost exactly, which confirms that the original study was carefully conducted. On new data, however, no spanning alpha remains significantly positive. The alpha of managed momentum is significantly smaller than in sample, and it remains so, narrowly, in a post-hoc adjustment for the smaller positions the strategy took, whereas the declines for profitability and value do not survive that adjustment. For an investor implementing the strategy in real time and paying trading costs, the point estimates of the Sharpe ratio gain are negative for the market, size, value, investment and profitability factors and close to zero for momentum, but every confidence interval also contains the gain the same strategy achieved before 2016. The primary pre-registered test, managed momentum at 10 basis points of costs, gave a gain of +0.03 with a 95% confidence interval from −0.32 to +0.39. With a power of only about 18% against a gain of the pre-2016 size, this test could not have been expected to settle the question. The evidence is therefore insufficient to conclude either that the strategy still works or that it has stopped working.

Beyond this specific strategy, the study illustrates how the design of a research project determines what its results can mean. A power analysis carried out before data collection showed that the most natural test, on the market factor, could not answer the question, and redirected the study towards momentum, although, as it turned out, only the momentum spanning test gained adequate power and the primary Sharpe ratio test did not. The lesson is that power must be computed for the exact statistic that will be tested. A pre-specified robustness grid revealed that 26 specifications, four distinct strategies, would have allowed a researcher to claim a significant out-of-sample success, and pre-registration is what allows those results to be reported as exploratory rather than as a discovery. A three-way decision rule prevented an underpowered null result from being presented as a refutation.

For practitioners, the results suggest caution towards volatility management applied factor by factor. Its historical advantage appears to have been concentrated in a few crash episodes, much of it could not have been captured in real time, and it did not appear in the decade after publication. For researchers, the main implication is methodological. Replicating a result is necessary but not sufficient: the question that matters is whether the result holds on data that were not used to discover it, under an analysis that was not shaped by the data. Future work could extend this protocol to the multifactor setting of DeMiguel et al. (2024), to international

factor data, and to longer variance windows, whose promising but exploratory results in this study now constitute a hypothesis that can be tested honestly on the next decade of returns.

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Appendixes

Appendix A. Out-of-sample spanning regressions, January 2016 to July 2026 (White robust SEs).

	Dependent variable:					
	Mkt^{σ}	SMB^{σ}	HML^{σ}	Mom^{σ}	RMW^{σ}	CMA^{σ}
	(1)	(2)	(3)	(4)	(5)	(6)
Unmanaged factor (β)	0.597^{***}	0.395^{***}	0.273^{***}	0.241^{***}	0.211^{***}	0.341^{***}
	(0.115)	(0.042)	(0.035)	(0.037)	(0.032)	(0.038)
Constant (α, %/yr)	1.476	−0.714	−1.110	0.514	0.359	−0.807
	(4.553)	(0.828)	(0.904)	(1.034)	(0.504)	(0.837)
Observations	127	127	127	127	127	127
R^2	0.357	0.652	0.593	0.507	0.524	0.536
Adjusted R^2	0.352	0.649	0.589	0.503	0.520	0.532

Note: Heteroscedasticity-consistent (White, 1980) standard errors in parentheses. c fixed on pre-2016 data; gross of trading costs. p-values here are two-sided. Significance levels: $^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$. Source: Author's own calculations.

Appendix B. Deviations from the pre-registration plan.

Date	Change	Reason	Results seen before?
24/09/2026	Primary test moved from the market factor to momentum	Power analysis: 29% power for the market factor against 88% for momentum	No; no data downloaded
24/09/2026	Replication window ends April 2015 instead of December 2015	The published observation counts (1,065; 1,060; 621) only fit a sample ending April 2015; out-of-sample window unchanged	No; decided from the published table before any regression

Note: No other change was made to the pre-registered tests after the plan was written. The analyses added after the results were known (Table 3, Panel B; Table 8; the percentile p-values in Appendix C) are labelled post hoc where they appear. The plan was not lodged with a public registry, so its timing cannot be independently verified (see Section 5.4). Source: Pre-registration plan, deviation log.

Appendix C. All 48 momentum specifications of the robustness grid.

Window (months)	Scaling	Cap	Cost (bps)	OOS gain	95% CI	p	p (pct.)	Pre-2016 gain	OOS turnover
1	1/vol	None	0	+0.10	[−0.09, 0.27]	0.121	0.152	+0.34	0.25
	1/vol	None	10	+0.07	[−0.12, 0.24]	0.208	0.235	+0.30	0.25
	1/vol	None	25	+0.03	[−0.17, 0.20]	0.381	0.404	+0.24	0.25
	1/vol	None	50	−0.05	[−0.26, 0.13]	0.672	0.700	+0.14	0.25
1	1/vol	1.5	0	+0.11	[−0.08, 0.27]	0.101	0.131	+0.20	0.23
	1/vol	1.5	10	+0.08	[−0.11, 0.25]	0.172	0.201	+0.19	0.23
	1/vol	1.5	25	+0.04	[−0.15, 0.21]	0.317	0.343	+0.17	0.23
	1/vol	1.5	50	−0.03	[−0.22, 0.14]	0.600	0.633	+0.15	0.23
1	1/var	None	0	+0.08	[−0.26, 0.44]	0.335	0.316	+0.19	0.22
	1/var	None	10	+0.03 †	[−0.32, 0.39]	0.447	0.436	+0.13	0.22
	1/var	None	25	−0.06	[−0.41, 0.32]	0.616	0.612	+0.03	0.22
	1/var	None	50	−0.19	[−0.58, 0.21]	0.827	0.829	−0.13	0.22

Window (months)	Scaling	Cap	Cost (bps)	OOS gain	95% CI	p	p (pct.)	Pre-2016 gain	OOS turnover
1	1/var	1.5	0	+0.09	[−0.26, 0.44]	0.327	0.308	+0.30	0.22
	1/var	1.5	10	+0.03	[−0.31, 0.40]	0.435	0.427	+0.27	0.22
	1/var	1.5	25	−0.05	[−0.41, 0.33]	0.607	0.602	+0.23	0.22
	1/var	1.5	50	−0.19	[−0.57, 0.22]	0.819	0.823	+0.16	0.22
3	1/vol	None	0	+0.16	[−0.01, 0.30]	0.010	0.032	+0.34	0.09
	1/vol	None	10	+0.15	[−0.02, 0.28]	0.015	0.043	+0.33	0.09
	1/vol	None	25	+0.13	[−0.04, 0.27]	0.026	0.063	+0.30	0.09
	1/vol	None	50	+0.10	[−0.07, 0.24]	0.063	0.121	+0.26	0.09
3	1/vol	1.5	0	+0.16	[−0.01, 0.30]	0.010	0.032	+0.21	0.09
	1/vol	1.5	10	+0.15	[−0.02, 0.28]	0.015	0.043	+0.21	0.09
	1/vol	1.5	25	+0.13	[−0.04, 0.27]	0.026	0.063	+0.20	0.09
	1/vol	1.5	50	+0.10	[−0.07, 0.24]	0.063	0.121	+0.19	0.09
3	1/var	None	0	+0.25	[−0.06, 0.50]	0.026	0.053	+0.29	0.09
	1/var	None	10	+0.23	[−0.08, 0.48]	0.040	0.072	+0.26	0.09
	1/var	None	25	+0.20	[−0.12, 0.45]	0.064	0.106	+0.22	0.09
	1/var	None	50	+0.14	[−0.18, 0.40]	0.140	0.184	+0.15	0.09
3	1/var	1.5	0	+0.25	[−0.06, 0.50]	0.026	0.053	+0.30	0.09
	1/var	1.5	10	+0.23	[−0.08, 0.48]	0.040	0.072	+0.29	0.09
	1/var	1.5	25	+0.20	[−0.12, 0.45]	0.064	0.106	+0.27	0.09
	1/var	1.5	50	+0.14	[−0.18, 0.40]	0.140	0.184	+0.24	0.09
6	1/vol	None	0	+0.19	[0.03, 0.35]	0.006	0.010	+0.35	0.05
	1/vol	None	10	+0.19	[0.02, 0.34]	0.008	0.013	+0.34	0.05
	1/vol	None	25	+0.18	[0.01, 0.33]	0.011	0.017	+0.33	0.05
	1/vol	None	50	+0.16	[0.00, 0.31]	0.018	0.029	+0.31	0.05
6	1/vol	1.5	0	+0.19	[0.03, 0.35]	0.006	0.010	+0.21	0.05
	1/vol	1.5	10	+0.19	[0.02, 0.34]	0.008	0.013	+0.20	0.05
	1/vol	1.5	25	+0.18	[0.01, 0.33]	0.011	0.017	+0.20	0.05
	1/vol	1.5	50	+0.16	[0.00, 0.31]	0.018	0.029	+0.19	0.05
6	1/var	None	0	+0.29	[0.02, 0.55]	0.016	0.017	+0.30	0.05
	1/var	None	10	+0.28	[0.01, 0.54]	0.019	0.022	+0.29	0.05
	1/var	None	25	+0.26	[−0.01, 0.52]	0.024	0.028	+0.26	0.05
	1/var	None	50	+0.23	[−0.04, 0.49]	0.040	0.046	+0.22	0.05
6	1/var	1.5	0	+0.29	[0.02, 0.55]	0.016	0.017	+0.30	0.05
	1/var	1.5	10	+0.28	[0.01, 0.54]	0.019	0.022	+0.30	0.05
	1/var	1.5	25	+0.26	[−0.01, 0.52]	0.024	0.028	+0.29	0.05
	1/var	1.5	50	+0.23	[−0.04, 0.49]	0.040	0.046	+0.27	0.05

Note: Real-time strategies; OOS = January 2016 to July 2026. p is the pre-registered uncorrected one-sided bootstrap p-value, computed from the centred bootstrap distribution. p (pct.) is the one-sided p-value consistent with the percentile interval, the share of bootstrap gains at or below zero (post hoc). The two differ when the bootstrap distribution is skewed. The 95% CI is two-sided and therefore corresponds to a one-sided test at 2.5%, not 5%. For three- and six-month windows the leverage cap never binds, so capped and uncapped rows are identical. † Pre-registered primary specification. Source: Author's own calculations.

Appendix D. Data files and SHA-256 checksums.

File	SHA-256 (first 32 characters)
F-F_Momentum_Factor_CSV.zip	7ee14e892b0f7044902fdbc4e25cfaf1

File	SHA-256 (first 32 characters)
F-F_Momentum_Factor_daily_CSV.zip	b039d986db27fc831bcf1f52ba213491
F-F_Research_Data_5_Factors_2x3_CSV.zip	b8653b411cc5e28917e7ef643bb42f6d
F-F_Research_Data_5_Factors_2x3_daily_CSV.zip	478350b8d60831351fbf754bc593fc01
F-F_Research_Data_Factors_CSV.zip	b840dba55d319f4818fc7300e65c52ef
F-F_Research_Data_Factors_daily_CSV.zip	1916d331c2c51d2aee3d00215897d2b8

Note: The full checksums, all code and all output files are provided with this paper. Every table and figure is reproduced by running the scripts 01 to 07 in order, in under one minute.